

## <MG-2009> Data Analysis for Business-II/ Statistical Inference

| Form number                   | COURSE OUTLINE/ DOCUMENT |                           |              |
|-------------------------------|--------------------------|---------------------------|--------------|
| QEC-ISOB-D1-CW 0.4            |                          |                           |              |
| COURSE INSTRUCTOR INFORMATION | Name                     | Asma Masood Malik         |              |
|                               | email ID                 | Asma.Masood@isb.nu.edu.pk | Room # A-315 |

| DEGREE INFORMATION | Program | Batch     | Section(s) | Semester |      |
|--------------------|---------|-----------|------------|----------|------|
|                    |         |           |            |          |      |
|                    | BA      | Fall 2023 | C          | Fall     | 2024 |

| COURSE INFORMATION       | Course Category<br>C- Core/ E-Elective |                 | Code                                                                                                           | Title                                | Credit hours                |
|--------------------------|----------------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------|--------------------------------------|-----------------------------|
|                          | C                                      |                 | MG2009                                                                                                         | Data Analysis for Business-II        | 3                           |
|                          | Prerequisite(s)                        |                 |                                                                                                                |                                      |                             |
|                          | TA Required (Yes/No)                   |                 | No. of TA(s)                                                                                                   | Brief Justification                  |                             |
|                          | Yes                                    |                 | 1                                                                                                              | For Checking Assignments and Quizzes |                             |
| TEXT BOOK(s) INFORMATION | Title of Book                          |                 | Basic Business Statistics Concepts and Applications                                                            |                                      | Edition<br>14 <sup>rd</sup> |
|                          | Author(s)                              |                 | Mark L. Berenson, David M. Levine, Kathryn A. Szabat, David F. Stephan                                         |                                      |                             |
|                          | Publisher                              |                 | Pearson                                                                                                        |                                      |                             |
| Reference Material (s)   | 1.                                     | Title of Book   | Introduction to Statistics                                                                                     |                                      |                             |
|                          |                                        | Imprint details | Ronald E. Walpole                                                                                              |                                      |                             |
|                          | 2.                                     | Title of Book   | Business Statistics                                                                                            |                                      |                             |
|                          |                                        | Imprint details | J.K Sharma, Pearson Education                                                                                  |                                      |                             |
|                          | 3.                                     | Title of Book   |                                                                                                                |                                      |                             |
|                          |                                        | Imprint details |                                                                                                                |                                      |                             |
|                          | Support Material (s)                   | a               | <b>Certification;</b><br>Statistical Analysis Fundamentals using Excel<br>Coursera Certification: 1 to 4 Weeks |                                      |                             |
|                          |                                        | b               | <b>AI integration:</b><br>For plan and controlling functions of this course                                    |                                      |                             |

|                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <b>Brief Description of Course:</b> | This course builds upon the foundation of business statistics and focuses on the principles and techniques of statistical inference. In this course, students learn how to make inferences and draw conclusions about population parameters based on sample data. The course covers estimation, hypothesis testing, and the application of various statistical methods to real-world business scenarios. It helps students learn advanced level research skills to better perform market research and analyze various forms of big data available in the firms. The statistical skills are taught with available real-time data through manipulation in advanced industry software. Most of the statistical skills of this course are also practiced in the lab. The course also aligns with the needs of the industry for data analysts. |
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#### Course Objectives:

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|----|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | Understand the fundamental principles and methods of statistical inference, emphasizing their application in business scenarios.                |
| 2. | Develop skills in point estimation and interval estimation techniques to accurately estimate population parameters using sample data.           |
| 3. | Apply hypothesis testing procedures to evaluate business hypotheses, make data-driven decisions, and assess the validity of statistical claims. |
| 4. | Analyze and interpret statistical results to draw meaningful business insights, supporting strategic decision-making.                           |
| 5. | Enhance critical thinking by evaluating statistical evidence, considering ethical implications, and ensuring integrity in data analysis.        |
| 6. | Utilize statistical tools and software to conduct analyses, interpret results, and apply findings to real-world business situations.            |

#### Learning Outcomes:

|    |                                                                                                                                              |
|----|----------------------------------------------------------------------------------------------------------------------------------------------|
| 1. | Understand the principles and methods of statistical inference presented in the textbook "Basic Statistics: Concepts and Applications."      |
| 2. | Apply estimation techniques from the textbook to estimate population parameters based on sample data.                                        |
| 3. | Conduct hypothesis tests using concepts and applications from the textbook to make informed decisions in business scenarios.                 |
| 4. | Analyze and interpret statistical results using concepts and applications from the textbook to draw meaningful conclusions.                  |
| 5. | Evaluate ethical considerations in statistical inference using examples and cases from the textbook and maintain integrity in data analysis. |

|       | PLO 1 | PLO 2 | PLO 3 | PLO 4 | PLO 5 | PLO 6 | PLO 7 | PLO 8 | PLO 9 | PLO 10 |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|
| CLO 1 |       |       | ✓     | ✓     |       |       |       |       |       |        |
| CLO 2 |       |       | ✓     | ✓     |       |       |       |       |       |        |
| CLO 3 |       |       | ✓     | ✓     |       |       |       |       |       |        |
| CLO 4 |       |       | ✓     | ✓     |       |       |       |       |       |        |
| CLO 5 |       |       |       |       |       |       |       |       | ✓     |        |

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**Courseware Structure: (Mark X where applies)**

| <i>Lecture<br/>(Lect)</i> | <i>Multimedia<br/>(MM)</i> | <i>Exercise<br/>(Exer)</i> | <i>IT Labs<br/>(Lab)</i> | <i>Case<br/>Studies<br/>(CAS)</i> | <i>Individual<br/>Assignment<br/>(Assign)</i> | <i>Group<br/>Presentation<br/>(G-Pres)</i> | <i>Quizzes</i> |
|---------------------------|----------------------------|----------------------------|--------------------------|-----------------------------------|-----------------------------------------------|--------------------------------------------|----------------|
| <b>X</b>                  | <b>X</b>                   | <b>X</b>                   |                          | <b>X</b>                          | <b>X</b>                                      | <b>X</b>                                   | <b>X</b>       |

**COURSE CONTENTS:**

| Weeks          | Contents/Topics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>**Courseware Events</b><br><br>(MM/ IT<br>Lab/Case Study/<br>Assignment/<br>Presentation<br>etc.)                                                                                                             |              |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <b>Week-01</b> | Introduction to Data Analysis and Statistical Inference.<br><b><u>Introduction to Sampling techniques and Sampling Distributions.</u></b><br>Probability and Non-Probability Sampling techniques. i.e. Simple Random sampling, Stratified random sampling, Systematic sampling, Purposive, convenience, Quota and Snowball sampling.                                                                                                                                                                                                                                                                   | <b><u>Chapter # 7</u></b><br><b><u>Assignment #1:</u></b><br><a href="https://www.coursera.org/learn/statistics-fundamentals-using-excel">https://www.coursera.org/learn/statistics-fundamentals-using-excel</a> | CLO 2, 4     |
| <b>Week-02</b> | <b><u>Sampling Distribution</u></b><br>Central limit theorem, Sampling Distribution about mean, Difference between mean, proportion and difference between proportions.                                                                                                                                                                                                                                                                                                                                                                                                                                | <b><u>Chapter # 7,8</u></b><br><br><b><u>Quiz # 1</u></b>                                                                                                                                                        | CLO 2, 3,4,5 |
| <b>Week-03</b> | <b><u>Estimation</u></b><br>Point estimates, Interval Estimation, Properties of Good Point Estimator. Confidence intervals for a population mean for large samples, Confidence intervals for a population mean for small samples, Confidence interval of a percentage, Calculation of sample size                                                                                                                                                                                                                                                                                                      | <b><u>Chapter # 8</u></b>                                                                                                                                                                                        | CLO 3,4,5    |
| <b>Week-04</b> | <b><u>Fundamentals of Hypothesis Testing: One-Sample Tests</u></b><br><b><u>Fundamentals of Hypothesis Testing</u></b><br>The Critical Value of the Test Statistic<br>Regions of Rejection and Non rejection<br>Risks in Decision-Making Using Hypothesis Testing<br><b><u>Z Test for the Mean (Population variance Known)</u></b><br>The Critical Value Approach<br>The p-Value Approach<br>Confidence Interval Estimation and Hypothesis Testing<br>Hypothesis Testing Using the Critical Value Approach<br>Hypothesis Testing Using the p-Value Approach<br>Z Test of Hypothesis for the Proportion | <b><u>Chapter # 9</u></b><br><b><u>Quiz # 2</u></b>                                                                                                                                                              | CLO 1,3,4,   |
| <b>Week-05</b> | <b><u>Fundamentals of Hypothesis Testing: One-Sample Tests</u></b><br><b><u>t Test of Hypothesis for the Mean (Population variance Unknown)</u></b><br>The Critical Value Approach, p-Value Approach<br>Checking the Normality Assumption<br><b><u>One-Tail Tests</u></b><br>The p-Value Approach, The Critical Value Approach<br><b><u>Potential Hypothesis Testing Pitfalls and Ethical Issues</u></b><br><b><u>Power of the Test</u></b>                                                                                                                                                            | <b><u>Chapter # 9</u></b><br><b><u>Assignment # 2</u></b>                                                                                                                                                        | CLO 1, 3,4,5 |

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| <b>Week-06</b>  | <b><u>Two-Sample Tests</u></b><br><b>Comparing the Proportions of Two Independent Populations</b><br>Z-Test for the Difference Between Two Proportions<br>Confidence Interval Estimate                                                                                                                                                                                                                                                                                                                                   | <b><u>Chapter # 10</u></b><br><b><u>Quiz # 3</u></b>       | CLO 3,4,5    |
| <b>Week-07</b>  | <b><u>Two-Sample Tests</u></b><br><b>Comparing the Means of Two Independent Populations</b><br>Pooled-Variance t-Test, Evaluating the Normality Assumption,<br>Confidence Interval Estimate<br>Separate-Variance t -Test<br><b>Comparing the Means of Two Related Populations</b><br>Paired t Test, Confidence Interval Estimate<br>Comparing the Proportions of Two Independent Populations, F Test for the Ratio of Two Variances<br>Effect Size<br><b>F Test for the Ratio of Two Variances</b><br><b>Effect Size</b> | <b><u>Chapter # 10</u></b><br><b><u>Assignment # 3</u></b> | CLO 1, 3,4,5 |
| <b>Week-08</b>  | <b><u>The Completely Randomized Design: One-Way ANOVA</u></b><br>Analyzing Variation in One-Way ANOVA<br>F Test for Differences Among More Than Two Means<br>One-Way ANOVA F Test Assumptions<br>Levene Test for Homogeneity of Variance<br>Multiple Comparisons:<br>The Analysis of Means (ANOM)                                                                                                                                                                                                                        | <b><u>Chapter # 11</u></b>                                 | CLO 3,4,5    |
| <b>Week-09</b>  | <b><u>Non Parametric Tests</u></b><br>Chi-Square and Nonparametric Tests<br>Assumptions of the chi-square test<br>Chi-Square Test for the independence of attributes                                                                                                                                                                                                                                                                                                                                                     | <b><u>Chapter # 11</u></b><br><b><u>Assignment #4</u></b>  | CLO 1, 3,4   |
| <b>Week-10</b>  | Chi-Square Test for Differences Among More Than Two Proportions<br>The Marascuilo Procedure,<br>The Analysis of Proportions (ANOP)<br>Chi-Square Test of Independence<br>Wilcoxon Rank Sum Test for Two Independent Populations                                                                                                                                                                                                                                                                                          | <b><u>Chapter # 12</u></b><br><b><u>Quiz # 4</u></b>       | CLO 1, 3,4,5 |
| <b>Week-11</b>  | Kruskal-Wallis Rank Test for the One-Way ANOVA<br>Assumptions of the Kruskal-Wallis Rank Test<br>McNamara Test for the Difference Between Two Proportions<br>Chi-Square Test for the Variance or Standard Deviation<br>Wilcoxon Signed Ranks Test for Two Related Populations<br>Friedman Rank Test for the Randomized Block Design                                                                                                                                                                                      | <b><u>Chapter # 12</u></b>                                 | CLO 1, 3,4,5 |
| <b>Week -12</b> | <b><u>Simple Linear Regression</u></b><br>Simple Linear Regression Models<br>Determining the Simple Linear Regression Equation<br>Measures of Variation, Assumptions of Regression                                                                                                                                                                                                                                                                                                                                       | <b><u>Chapter # 13</u></b>                                 | CLO 1, 4, 5  |
| <b>Week 13</b>  | Inferences About the Slope and Correlation Coefficient<br>Estimation of Mean Values and Prediction of Individual Values, Potential Pitfalls in Regression                                                                                                                                                                                                                                                                                                                                                                | <b><u>Chapter # 13</u></b><br><b><u>Assignment #5</u></b>  | CLO 2, 4, 5  |
| <b>Week-14</b>  | <b><u>Time-Series Forecasting</u></b><br>Time-Series Component Factors<br>Smoothing an Annual Time Series                                                                                                                                                                                                                                                                                                                                                                                                                | <b><u>Chapter # 15</u></b>                                 | CLO 3, 4, 5  |

|                |                                                                                                          |                     |             |
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|                | Least-Squares Trend Fitting and Forecasting<br>Autoregressive Modeling for Trend Fitting and Forecasting | <b>Quiz # 5</b>     |             |
| <b>Week-15</b> | <b>Choosing an Appropriate Forecasting Model</b><br>Time-Series Forecasting of<br>Seasonal Data          | <b>Chapter # 16</b> | CLO 3, 4, 5 |
| <b>Week-16</b> | <b>Revision of Syllabus</b>                                                                              |                     |             |

### Grading Criteria

|   |                         |                         |
|---|-------------------------|-------------------------|
| ■ | <b>ABSOLUTE Grading</b> | <b>RELATIVE Grading</b> |
|---|-------------------------|-------------------------|

#### Marks Distribution:

| Particulars                        | % Marks                | *Weight Ranges         |
|------------------------------------|------------------------|------------------------|
| 1. Interactive Session assignments | 10                     | 5 ~ 10                 |
| 2. Pop Quizzes                     | 10                     | 5 ~ 10                 |
| 3. Sessional – I                   | 15                     | 10 ~ 20                |
| 4. Sessional – II                  | 15                     | 10 ~ 20                |
| 5. Final Exam                      | 50                     | 40 ~ 50                |
| <b>Total:-</b>                     | <b>100</b>             | <b>100</b>             |
| Particulars                        | Planned (Qz/A s/Lab s) | Remarks                |
| 1. Quizzes                         | 4                      | quizzes announced      |
| 2. Assignments/                    | 4                      | Individual assignments |

#### Planned Courseware Events:

\*Weight *Ranges as defined in Flex*

|                              |                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>QUALIFYING ATTENDANCE</b> | <p>You must attend every class for your own personal benefit. Please refer to university policy of minimum attendance requirement (<u>currently 6 absentees – 1.30 hrs (90 minutes) each class duration</u>).</p> <p>Failing to conform qualifying attendance threshold, the student will stand debarred from sitting in the examination and assigned with “F” Grade.</p> |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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| <b>Academic and Moral Integrity:</b> | <b>All assignments should be your own work (or your group's when approved). PLAGIARISM will be awarded with "F" grade and/or reported to the University for academic and moral misconduct. To protect yourself, ALWAYS PROVIDE REFERENCES!</b> |
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### Instructions / Suggestions for STUDENTS for satisfactory progress in this course:

- ✓ FAST has a 100 percent attendance policy. However, students are allowed to sit in the final exam if they have at least 80 percent attendance.
- ✓ Do not schedule other appointments during your lecture time, still If you miss a lecture; Ask for a copy of the lecture notes. Get a good, readable and reliable copy and work though it carefully and diligently.
- ✓ Plagiarism in any part of the course can lead to an **F** in the whole course.
- ✓ On average, most students should find at least three hours outside of class for each class hour for satisfactory learning.
- ✓ **Note:** Counseling hours are displayed outside the doors of permanent faculty members for providing extra guidance to students. However, students are welcome to get help from the instructor at all times.
- ✓ If you encounter academic difficulties in your course, do not wait, contact your instructor or advisor in designated counseling hours and seek appropriate levels of assistance to help resolve your academic apprehensions.
- ✓ Inform your course instructor, advisor, administrator or staff person of your concerns and provide them with all relevant information on the challenges you are facing.
- ✓ Read the objectives and learning outcomes well; students are expected to attain the intended learning outcomes.
- ✓ Plan all of your activities; use the calendars, things to do list, reminders or other resources to help you keep track of important tasks, course assignments, quizzes, examination dates and deadline dates as well as your co-curricular activities.
- ✓ Students unable to submit an assignment by the due date should discuss the matter in advance with the course instructor. At the instructor's discretion, late assignments may not be accepted. Where late assignments are accepted, a late penalty of 10% per day will be levied.
- ✓ Assignments/ Activities: They are not meant simply for grades, but to reinforce your learning. Assignments are due on time. Each day late will lower your assignment grade by 10%. Apart from value of content, spelling, grammar, punctuation, and good presentation (printing and paper quality) will figure into your assignment grade.
- ✓ Copying other students' work or allowing your own work to be copied, both are considered cheating.
- ✓ You must log in to your SLATE account at least once in every 24 hours for announcements, assignments, lecture notes and messages
- ✓ Check FLEX regularly for your attendance, and evaluations.
- ✓ You need to enter the lecture room fully awake and able to devote your best concentration, for this go to bed at reasonable times and take optimum sleep.
- ✓ Come to each lecture properly prepared, bring in last lectures' notes, textbooks, ball points/pencils, laptops and calculators etc.
- ✓ Use the few minutes you usually have before the start of each class to review the prior lectures' notes and homework. This will save us valuable in-class time to work on new material.
- ✓ Chapters should be read and homework/exercises should be attempted before class.
- ✓ Do not get behind. You are encouraged to work with other students.
- ✓ The homework assigned is a minimum. You should always work extra hours on your own.
- ✓ Develop a learning habit rather than memorizing; work in groups, whenever appropriate.
- ✓ Apply the learned principles and gained knowledge; be creative in thinking.
- ✓ To guard against errors, please keep copies of the papers you turn in and retain all graded assignments for your reference.
- ✓ It is the responsibility and duty of students to become acquainted and abide by all the policies and rules of the FSB.

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**Note:** Students are always welcome to get help from the Teacher.